

Ananya Giliyal

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PROFESSIONAL SUMMARY

Data Scientist specializing in **GenAI, LLMs, Computer Vision, and NLP**. Built **COMET**, an event-driven LLM orchestration platform, and delivered **15+ production AI systems** that reduced vendor costs by **70%** and automated large-scale media workflows.

EXPERIENCE

JioStar India Pvt Ltd (formerly Viacom18 Media)

Jan 2024 – Present

Assistant Manager, Data Science & Analytics

Mumbai, India

- Architected **COMET**, an event-driven AI orchestration platform for JioHotstar's content operations team, deploying **6+ parallel LLM microservices** for speech-to-text, emotion extraction, multilingual translation, dubbing, face clustering, and scene/object/OCR/brand/smoke detection on GPU/CPU clusters. Built a central vector database storing timecode-level metadata powering downstream AI services through Contido MAM APIs.
- Built a **broadcast-grade Subtitle Generation Pipeline** on Whisper STT with dual generation routes (offline model + LLM), producing subtitles for **3,000+ videos** aired across JioStar properties; currently fine-tuning for low-resource Indian languages.
- Designed and deployed **Short-Form Video & Clip Generation** pipelines (9:16 / 16:9) using ASD, OCR, Autoflip, face recognition, object tracking, silence detection, and LLM transcript analysis across IPL (2024–26), WPL, GEC and animation. Achieved **2M+ views** with **<3-minute turnaround** from live event to publish.
- Built an automated **Promo Generation System** combining scene descriptors, speaker diarisation (NeMo, Pyannote), camera shot detection, face recognition and audio highlight detection, reducing manual effort by **150+ hours/month**.
- Developed an end-to-end **AI Dubbing Pipeline** with speaker diarisation and dynamic model routing across Vertex AI, OpenAI and ElevenLabs for multilingual content generation.
- Built an **Ad Break Prediction** system using VAD, silence detection, pixel-change analysis and transcription signals to identify natural commercial insertion points.
- Reduced external vendor costs by **70%** by replacing third-party subtitling, dubbing and content-tagging workflows with in-house AI pipelines.
- Developed **6+ internal Streamlit applications** including a Sprinklr-based social listening platform with an LLM insights layer, MST broadcast scheduling (BARC analysis and rating prediction), and BIBO viewer retention modelling for Star Utsav and Star Maa.
- Collaborated with MTS, Sports (IPL/WPL), Marketing (MTV, Star, Digital), Strategy and Digital Ventures teams while mentoring interns and coordinating production AI deployments.

Dun & Bradstreet

Jan 2023 – Jun. 2023

Data Science Intern

Mumbai, India

- Delivered **market segmentation and risk profiling** solutions for B2B clients using micro-segmentation, clustering, and look-alike modelling, identifying high-potential importer-exporter matches across **20+ countries** by product and service category.
- Built scalable **PySpark pipelines** for processing and transforming large shipping and vendor datasets, enabling efficient real-time analytics and business insights.

PROJECTS

netcdf-unpacker

Feb 2024

Python, NVIDIA RAPIDS, Geospatial Data Processing

- Published an open-source Python library on **PyPI** for unpacking and processing large-scale geospatial NetCDF datasets including **LANDSAT, MODIS, and GridMET** (50GB+).
- Accelerated large-scale geospatial data processing using **NVIDIA RAPIDS**, enabling GPU-optimised workflows for crop-yield prediction research.

EDUCATION

Mukesh Patel School of Technology, Management & Engineering, NMIMS

2020 – 2024

B.Tech in Data Science | GPA: 3.94/4.0 | Overall Rank 1 - Dean's List

Mumbai, India

SKILLS

Languages: Python, SQL, Shell Scripting

GenAI & LLMs: OpenAI, Vertex AI (Gemini), Whisper, Hugging Face Transformers, ElevenLabs, RAG, LLM Routing, Fine-tuning, Vector Databases, Sentence Transformers

Vision models, NLP & Audio: YOLO, OpenCV, FFmpeg, PaddleOCR, Tesseract, MediaPipe, MTCNN, MoviePy, Face Clustering, Scene/Object Detection, Multilingual NLP, Speaker Diarisation (NeMo, Pyannote), spaCy, NLTK, Librosa

Data & Tools: PySpark, Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Docker, AWS (S3, EC2, Lambda), Git, GPU/CPU Clusters, Streamlit, FastAPI, Tableau, Jupyter